



SUPPLY CHAIN ANALYTICS & PROFITABILITY OPTIMIZATION

DataCo's Supply Chain Case Study And Analytics

An End-to-End Business Intelligence and
Supply Chain Analytics Project

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BUSINESS PERFORMANCE OVERVIEW

This dashboard summarizes DataCo's business performance from 2015–2017. The company generated \$36.78M in revenue and \$3.97M in profit, achieving an 11% profit margin. Sales remained stable during 2015–2016 before declining sharply in Q4 2017, indicating changing demand patterns that require further investigation.

BUSINESS INSIGHTS - DATACO ECOMMERCE

TOTAL PROFIT	TOTAL REVENUE	PROFIT MARGIN	TOTAL ORDERS	CUSTOMERS
3,966,903	36,784,735	11	384,079	20,652

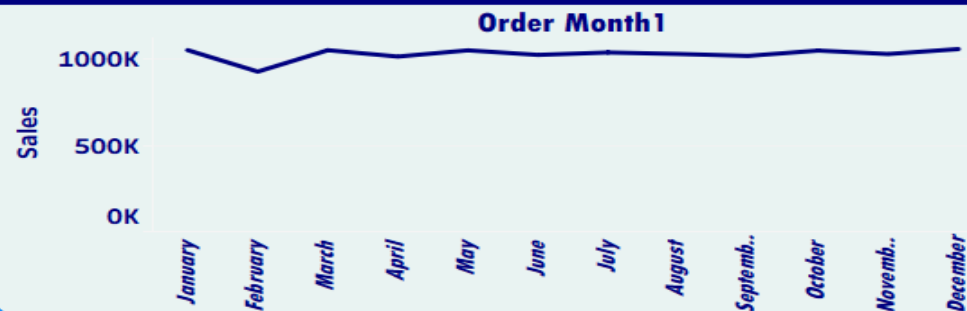
SALES TREND BY YEARS

2017



- Revenue declined significantly during Q4 2017.

2015



2016



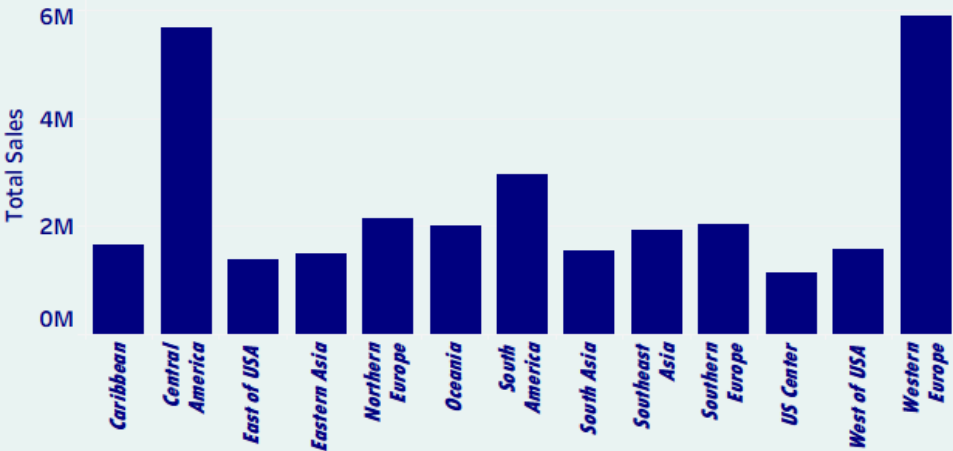
REVENUE AND PROFITABILITY ANALYSIS

Revenue and profitability differ significantly across markets and product categories. Central America and Western Europe contribute the highest revenue and profit, while several categories generate comparatively low returns. Understanding these variations helps prioritize investments and improve profitability.

REVENUE AND PROFIT OVERVIEW

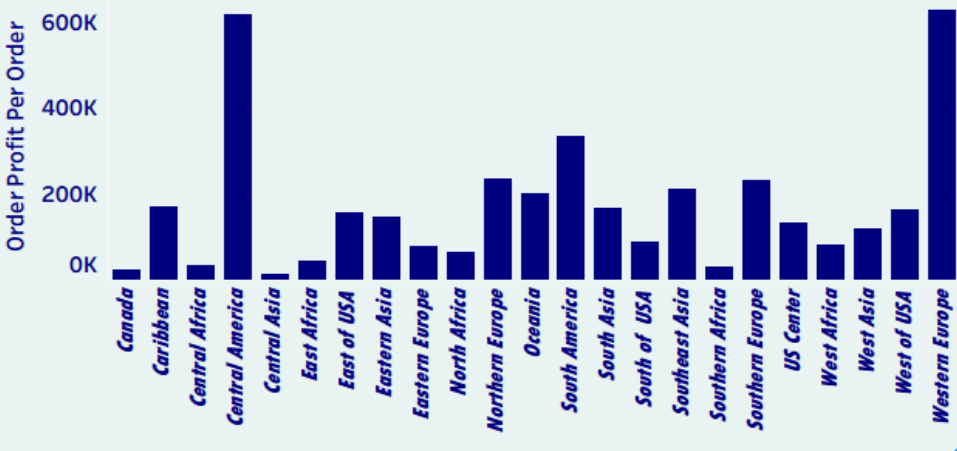
SALES BY REGION

Order Region



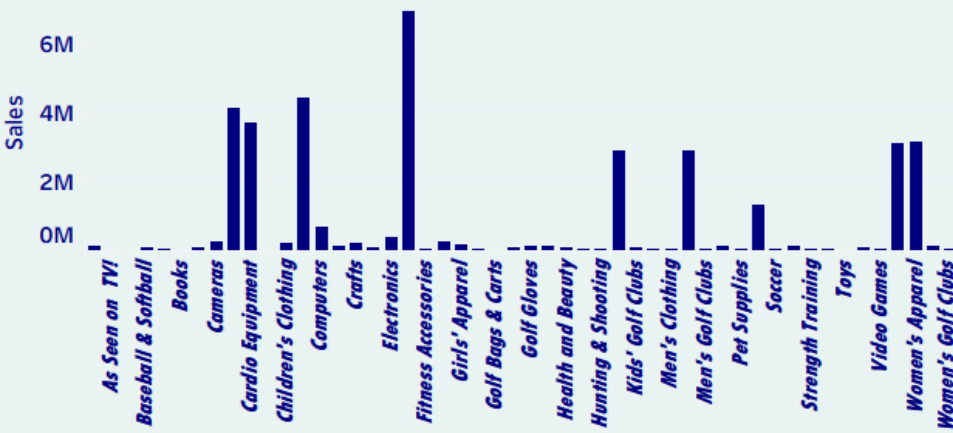
Profit By Region

Order Region



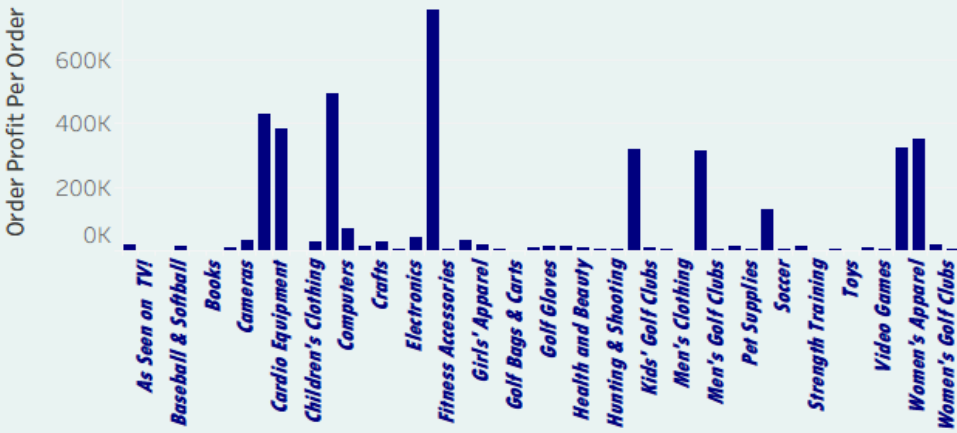
Sales By Category

Category Name



Profit By Category

Category Name



Business Recommendation

- Focus inventory on profitable regions.
- Expand high-margin categories.
- Review underperforming categories.

PRODUCT ANALYSIS

Product-level analysis reveals the highest-performing and lowest-performing products while evaluating the relationship between discounts and profitability. The results help identify products that should receive inventory priority and those requiring pricing strategy adjustments.

PRODUCT ANALYSIS			
TOP PRODUCTS		LOW-SALES PRODUCTS	
Product Name		Product Name	
Field & Stream Sportsman 16 Gun Fi..	6,929,654	Polar Loop Activity Tracker	5,937
Perfect Fitness Perfect Rip Deck	4,421,143	Bowflex SelectTech 1090 Dumb..	6,000
Diamondback Women's Serene Class..	4,118,426	Merrell Women's Grassbow Spo..	7,539
Nike Men's Free 5.0+ Running Shoe	3,667,633	Nike Men's Fingertrap Max Tra..	7,999
Pelican Sunstream 100 Kayak	3,099,845	LIJA Women's Button Golf Dress	8,208
Nike Men's CJ Elite 2 TD Football Cl..	2,891,758	Cleveland Golf Women's 588 RT..	8,399
Under Armour Girls' Toddler Spine S..	1,269,083	Diamondback Girls' Clarity 24 ..	8,400
Dell Laptop	663,000	Diamondback Boys' Insight 24 ..	8,700
Web Camera	267,608	TaylorMade Women's RBZ SL R..	8,839
Lawn mower	257,769	Stiga Master Series ST3100 Co..	8,910



- ### Business Recommendation
- Prioritize inventory for top-performing products.
 - Reduce excessive discounts.
 - Consider discontinuing consistently low-performing products.

LOGISTICS OVERVIEW

Logistics performance directly affects customer satisfaction and operational efficiency. The analysis evaluates delivery delays, shipping methods, and late-delivery rates to identify opportunities for improving supply chain performance.

LOGISTICS OVERVIEW

Avg. Delivery Delay

0.57

Late Delivery%

54.83

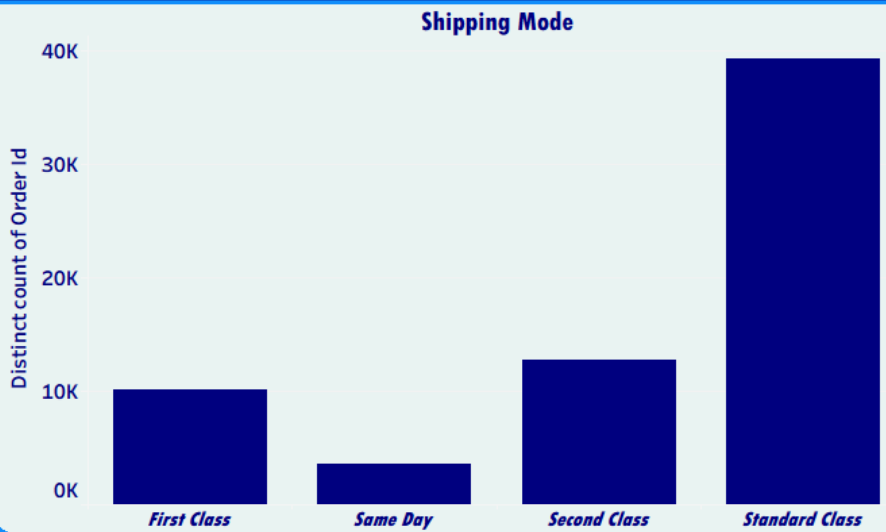
On Time Delivery Rate

42.72

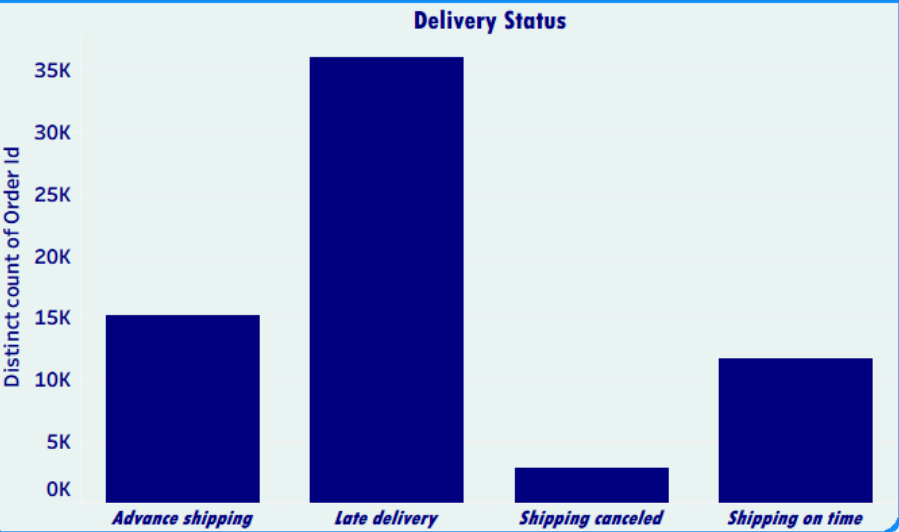
KEY INSIGHT

Standard Shipping processes the largest number of orders.

Orders vs Shipping Mode



Orders vs Delivery status



Business

Recommendation

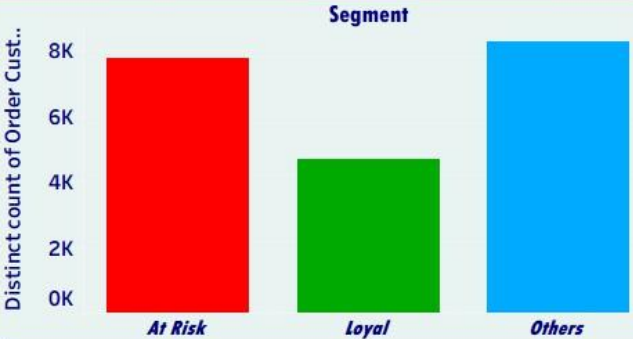
- Improve Standard Shipping efficiency.
- Reduce late deliveries.
- Optimize shipping mode allocation.

Customer Segmentation & Retention

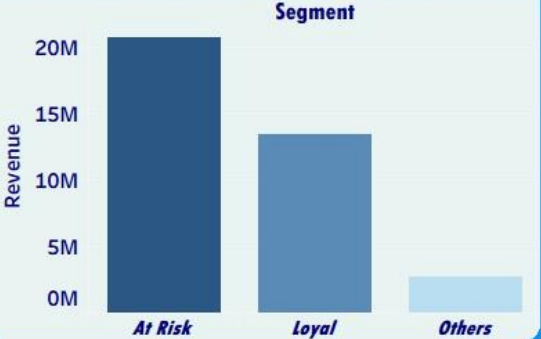
Customer segmentation using RFM Analysis identifies valuable customer groups and opportunities for retention. Cohort analysis measures repeat purchasing behavior and helps evaluate long-term customer loyalty.

Customer Dashboard

Customers vs Segments



Revenue vs Segment



Top Customers

Customer Name	
Christopher Smith	85,271
Mary Jones	88,860
Elizabeth Smith	90,516
Michael Smith	92,843
William Smith	108,523
John Smith	129,883
David Smith	145,777
James Smith	147,924
Robert Smith	152,231
Mary Smith	4,771,646

Business Recommendation

- Reward loyal customers.
- Launch retention campaigns for at-risk customers.
- Increase marketing toward high-value customer segments.

COHORT HEATMAP

Retention Rate
0.0031 0.2545

Cohort M...	Cohort Index																						
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
2015-01	0.1145	0.1355	0.1289	0.1247	0.1397	0.1175	0.1217	0.1343	0.1343	0.1241	0.1361	0.1235	0.1187	0.1289	0.1289	0.1529	0.1217	0.1463	0.1319	0.1427	0.1241	0.1319	0.1355
2015-02	0.1398	0.1283	0.1268	0.1230	0.1360	0.1413	0.1253	0.1375	0.1306	0.1451	0.1306	0.1238	0.1490	0.1131	0.1245	0.1352	0.1169	0.1398	0.1245	0.1360	0.1360	0.1367	0.1261
2015-03	0.1285	0.1357	0.1277	0.1325	0.1349	0.1325	0.1438	0.1060	0.1237	0.1414	0.1221	0.1349	0.1365	0.1301	0.1317	0.1309	0.1189	0.1205	0.1430	0.1189	0.1406	0.1245	0.1269
2015-04	0.1349	0.1453	0.1358	0.1235	0.1320	0.1339	0.1263	0.1349	0.1349	0.1206	0.1216	0.1358	0.1415	0.1292	0.1349	0.1320	0.1263	0.1358	0.1406	0.1292	0.1235	0.1187	0.1358
2015-05	0.1383	0.1414	0.1125	0.1280	0.1238	0.1445	0.1517	0.1249	0.1290	0.1197	0.1311	0.1166	0.1125	0.1383	0.1259	0.1300	0.1331	0.1187	0.1445	0.1404	0.0960	0.1280	0.1414
2015-06	0.1302	0.1443	0.1482	0.1108	0.1392	0.1521	0.1443	0.1211	0.1418	0.1186	0.1276	0.1392	0.1263	0.1534	0.1392	0.1289	0.1353	0.1379	0.1418	0.1211	0.1366	0.1495	0.1327
2015-07	0.1427	0.1201	0.1370	0.1158	0.1116	0.1483	0.1229	0.1201	0.1299	0.1525	0.1299	0.1285	0.1299	0.1314	0.1455	0.1243	0.1271	0.1144	0.1314	0.1299	0.1229	0.1370	0.1201
2015-08	0.1308	0.1199	0.1262	0.1371	0.1137	0.1044	0.1308	0.1044	0.1402	0.1355	0.1293	0.1137	0.1542	0.1449	0.1417	0.1153	0.1433	0.1168	0.1340	0.1293	0.1480	0.1340	0.1417
2015-09	0.1220	0.1201	0.1417	0.1220	0.1299	0.1496	0.1535	0.1280	0.1240	0.1102	0.1358	0.1260	0.1673	0.1142	0.1299	0.1339	0.1181	0.1417	0.1398	0.1555	0.1339	0.1201	0.1358
2015-10	0.1322	0.1279	0.1322	0.1109	0.1365	0.1322	0.1130	0.1194	0.1279	0.1535	0.1023	0.1322	0.1215	0.1215	0.1194	0.1130	0.1450	0.1279	0.1365	0.1493	0.1407	0.1237	0.1471
2015-11	0.1194	0.1475	0.1452	0.1148	0.1124	0.0984	0.1194	0.1499	0.1429	0.0984	0.1077	0.1358	0.1358	0.1265	0.1194	0.1382	0.1171	0.1218	0.1194	0.1218	0.1265	0.1171	0.0094
2015-12	0.1148	0.1401	0.1232	0.1597	0.1513	0.1176	0.1401	0.1148	0.1597	0.1148	0.1008	0.1289	0.1232	0.1036	0.1401	0.1485	0.1232	0.1148	0.1457	0.1709	0.1709	0.0084	
2016-01	0.0939	0.1586	0.1715	0.1359	0.0874	0.1230	0.1683	0.1100	0.1294	0.1327	0.1197	0.1327	0.1100	0.2006	0.1262	0.1359	0.1294	0.1003	0.1294	0.1100	0.0065		
2016-02	0.1535	0.1339	0.1299	0.1102	0.1063	0.1496	0.1181	0.1181	0.1299	0.1024	0.1299	0.1575	0.1299	0.0787	0.0984	0.1142	0.1575	0.1417	0.1142	0.0039			

Demand Forecasting & Supply Chain Optimization

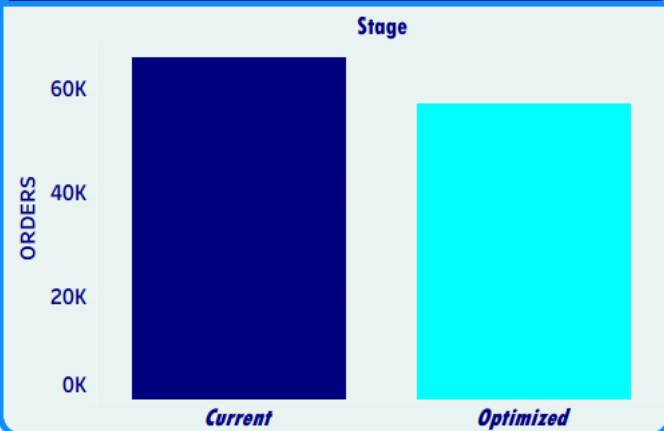
Demand forecasting predicts future sales trends, while optimization models prioritize products and markets using linear programming. Scenario analysis demonstrates how inventory constraints and market prioritization improve strategic decision-making under limited resources.

Forecasting And Optimization

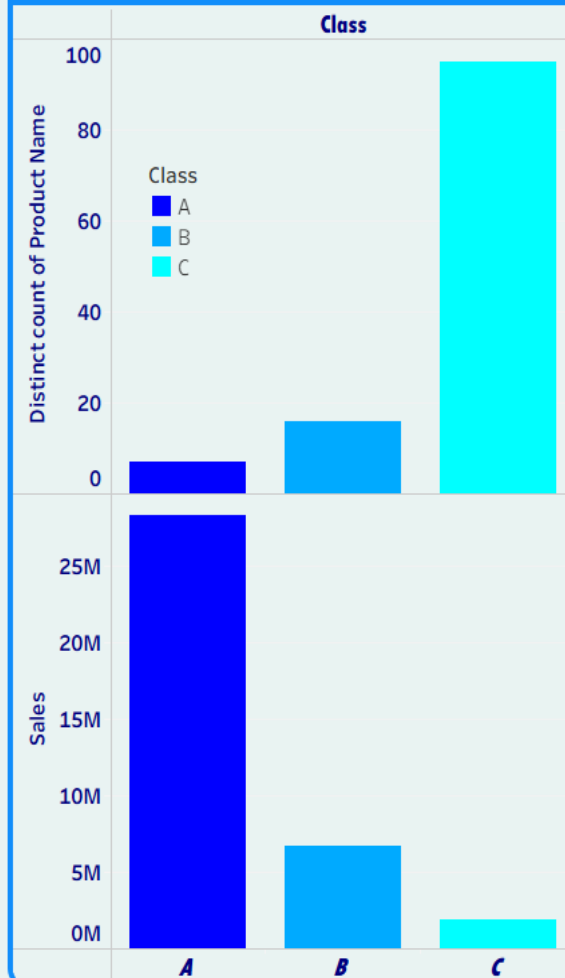
Actual vs Forecast Sales



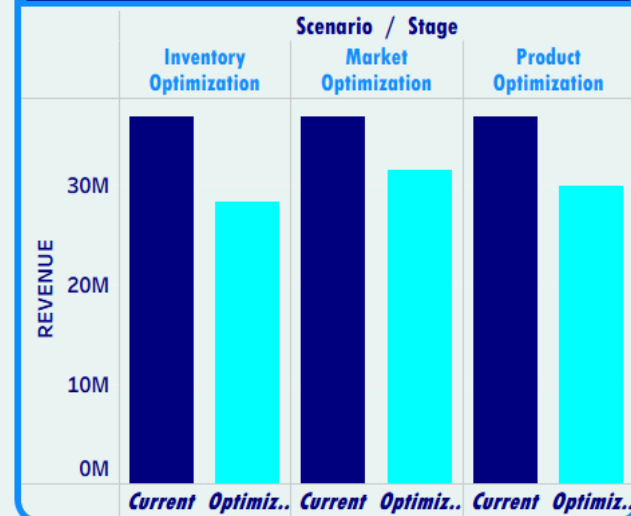
Orders vs Stage(in market optimization)



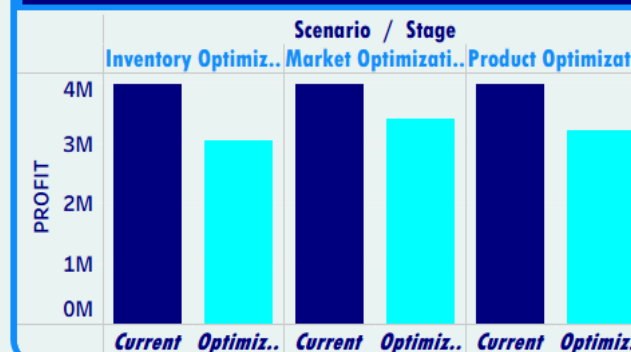
Class vs SALES and Products



Revenue vs Stage



Profit vs Stage



Key Insights

- ABC Analysis identifies Class A products for inventory prioritization.
- Product optimization improves inventory allocation.
- Market optimization recommends prioritizing high-performing markets.
- Scenario analysis supports strategic planning under different business conditions.

Business Recommendations

- Increase inventory for Class A products.
- Prioritize markets selected by the optimization model.
- Use demand forecasts for procurement planning.
- Improve logistics in high-delay markets.
- Continuously monitor profitability through executive dashboards.